What Does "Linear" Mean?

Beginner

Before the algorithm, the shape: what a straight-line relationship looks like, and what it doesn't.

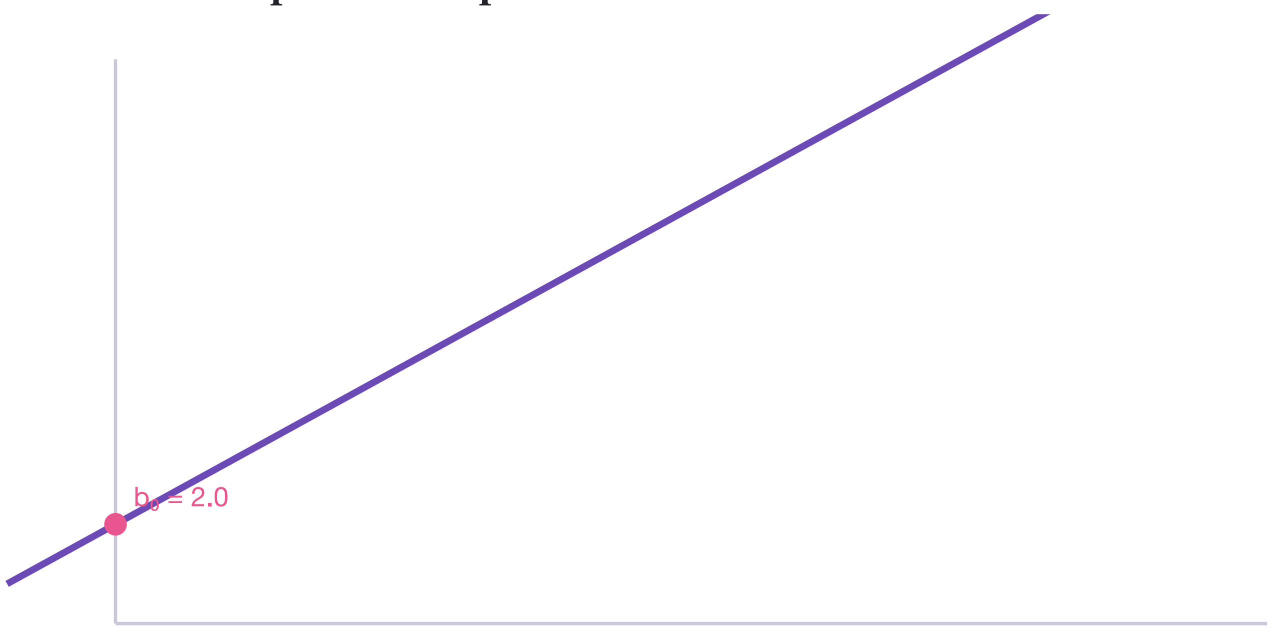
THE SIMPLEST FORM

$$y = b_0 + b_1 x$$

x is the input, y is the output, b_0 is the **intercept** (where the line crosses the y-axis, when x = 0), and b_1 is the **slope** (how steeply y changes as x increases).

"Linear" simply means: as x increases by a fixed amount, y changes by a roughly *constant* amount, every time — not more at first and less later, not curving. That's the geometric picture of a straight line.

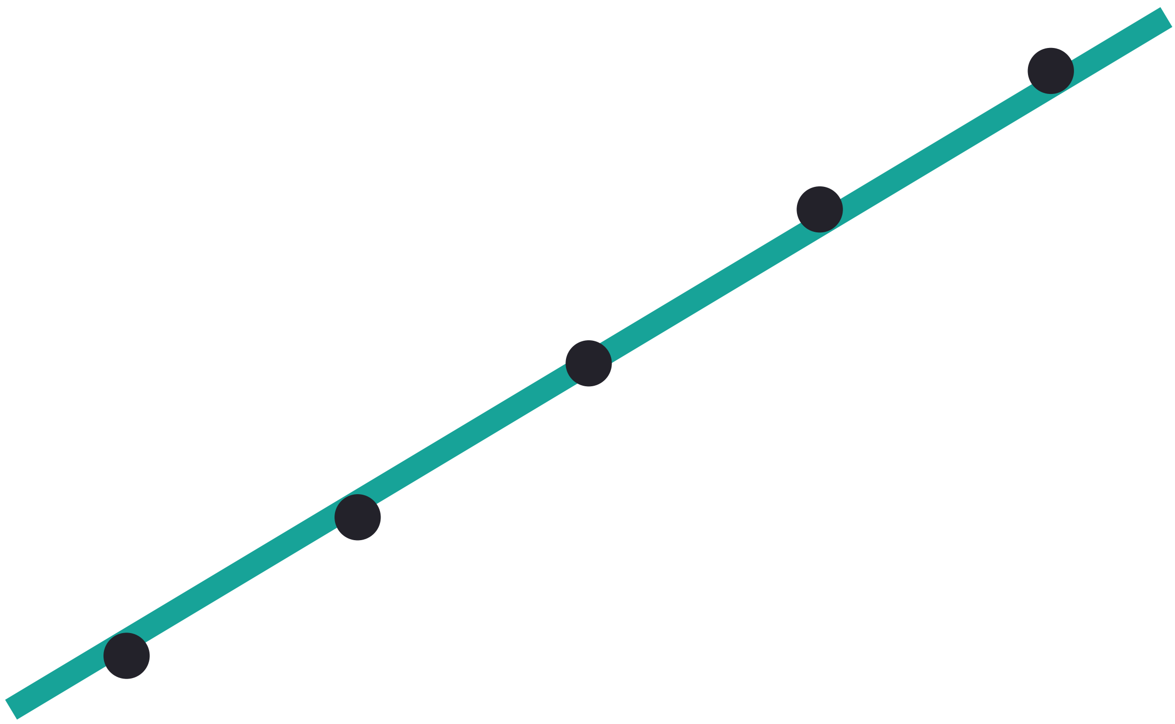
Interactive — Slope & Intercept



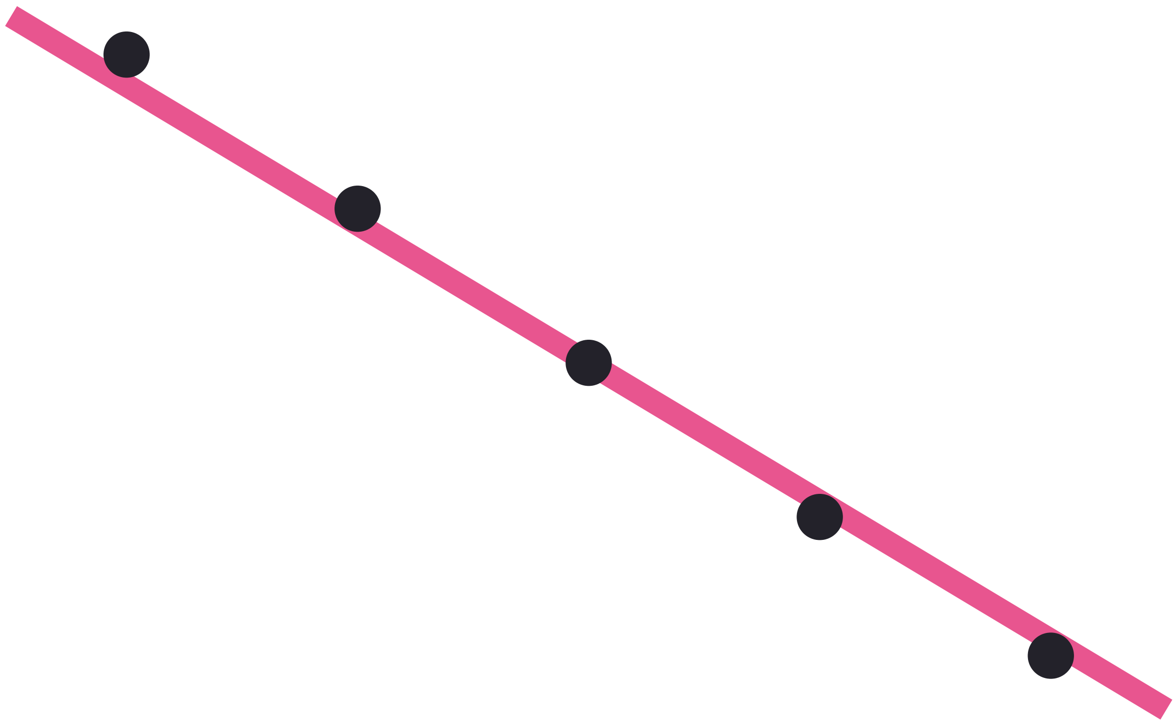
Equation: $\hat{y} = 2.0 + (1.2)\cdot x$

Drag the sliders. Notice: b_0 only ever moves where the line crosses the y-axis; b_1 changes the steepness (and its sign flips the tilt).

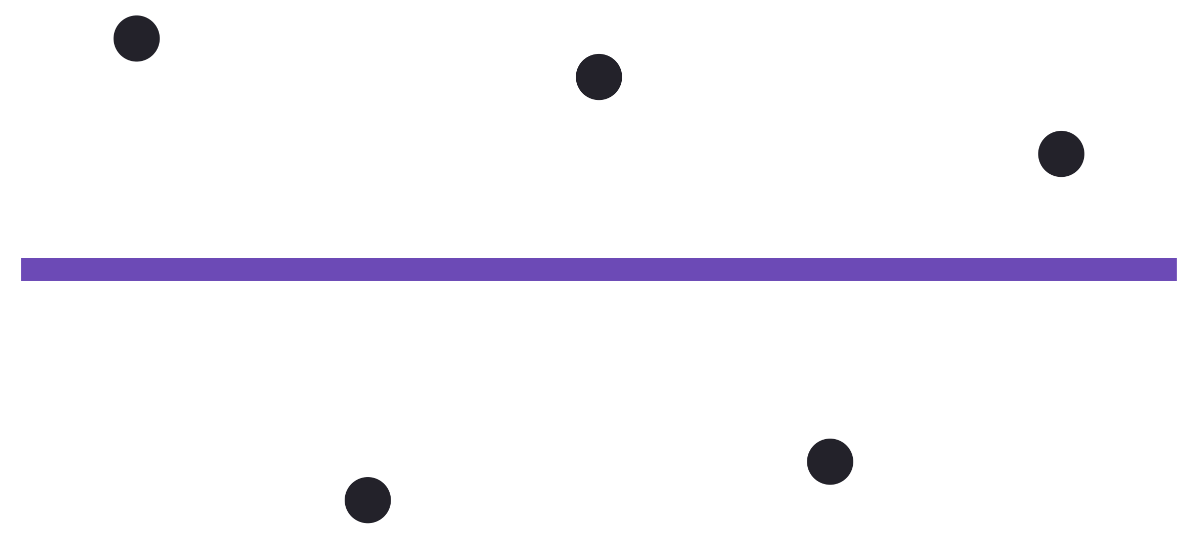
Positive, negative, and (approximately) no relationship



Positive slope: y rises as x rises



Negative slope: y falls as x rises



Near-zero slope: little linear relationship

COMMON MISTAKE

Assuming that because two variables aren't linearly related, they aren't related *at all*. A U-shaped or cyclical relationship can be strong while its linear correlation is near zero (Chapter 22 covers how to still use linear regression on curved data).